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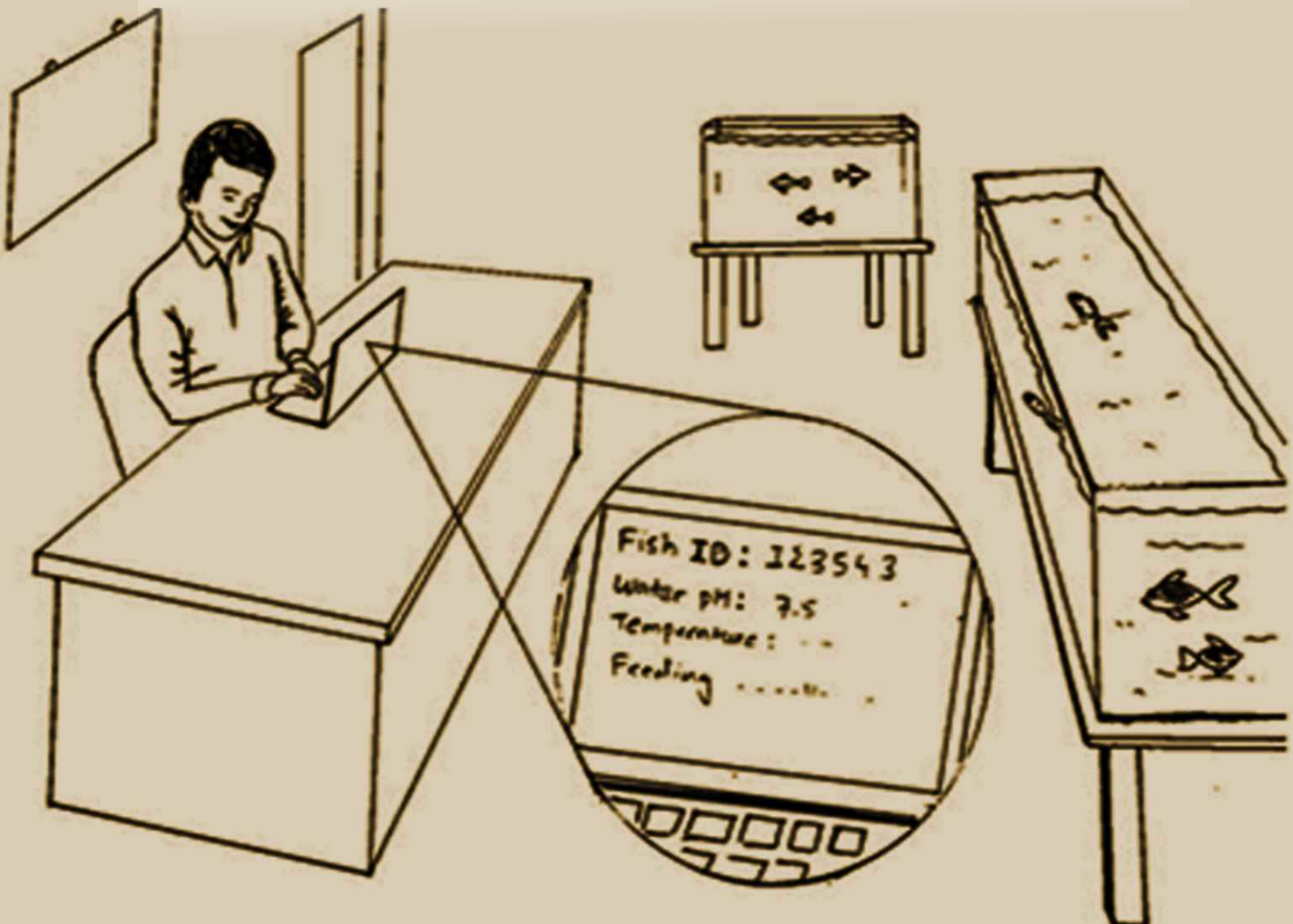
Chapter 12

Record keeping at ornamental fish production & marketing establishments

V.P. Saini

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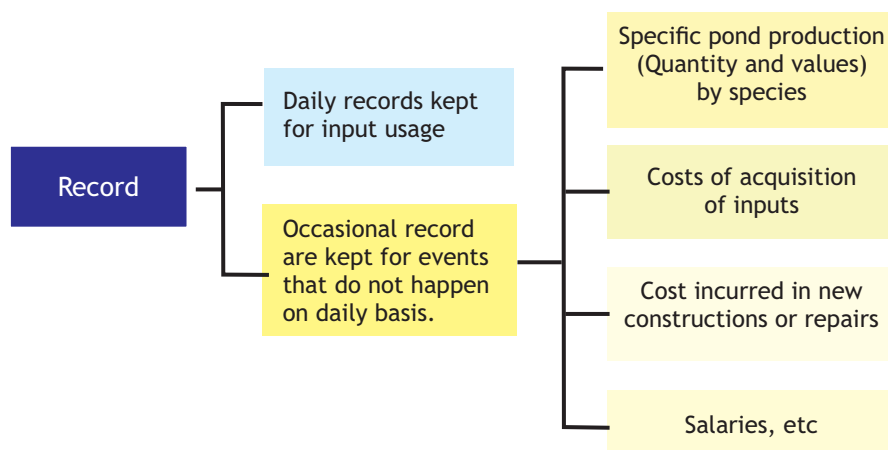
1. An overview

Record keeping facilitates smooth functioning and reduces risk for an enterprise

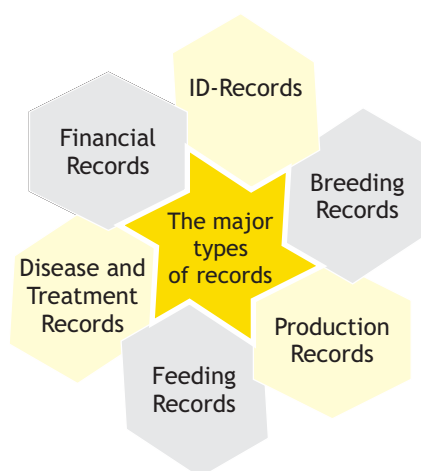
One of the main drawbacks on the economic operation of ornamental fish investment in India is a lack of deliberately kept economic records of production operations. Investors who operate without records are likely to make wrong decisions due to lack of information of what is happening at their farms. The best sources of information needed to advice on proper running of ornamental fish farm are properly designed and kept farm records. Record is information that has been systematically and carefully collected and appropriately stored for intended use. In order to run any business successfully, carefully thought out, properly collected and kept records are a must. For the purpose of keeping track and decision making in ornamental fish business, comprehensive and well maintained records must be kept. Just as in any other enterprise, properly collected and kept records are also important in ornamental fish farming. The properly maintained records can be used:

- to determine profitability of various techniques of production
- to compare the efficiency of use of inputs, such as land, labour and capital, with that of alternative production activities
- to help the investor in improving the efficiency of farm's operations
- to preserve institutional memory of the enterprise for future reference.
- to forecast the production
- to determine the amount of inputs requirements for specific activity
- to determine the financial health of the enterprise

Classification of Ornamental Fish Farm Records

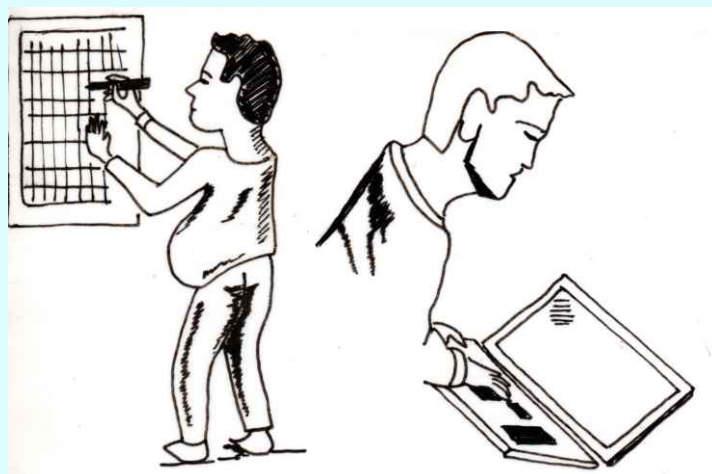


Types of Records



Farm record keeping methods

Farm record keeping methods range from the simple manual record (notebook & logbook) to sophisticated computer accounting systems. The manual farm record book remains the old stand-by for farm record keeping due to its ease of use while the computer accounting systems vary in complexity and need for technical assistance. Many of the programmes available require a minimum of computer knowledge yet offer fast and accurate management records. There are numerous kinds of farm record keeping systems available in the market. For example, TALLY & QUICKEN computer software are widely used to keep financial records and Microsoft excel for input output records. Review the different options before finalizing on the right system for your operation and decide for one that fits your specific farm operation. It is important to keep farm records as simple as possible but to record all necessary details in order that the performance of the farm operation can be fully evaluated.



2. BMP-1: Keep Identification (ID) records for tracking farm facilities and fish stock



The identification records of farm facilities and fish brood stock are essential for farm manager to track the production, sale, breeding performance of individual fish and to give instructions to farm workers. Giving the fish name/number and keeping a table with the characteristics of the fish and link it to the name can work in many cases. The methods of identification can be subdivided into 2 categories: internal (PIT) and external (fin clipping, paint and dye, fin tagging, etc.).

Guiding factors for BMP-1

- 2.1 Prepare a detailed layout plan for all the facilities (i.e. ponds, buildings, net houses, etc.) available at farm
- 2.2 Give a unique ID number to each facility
- 2.3 The ID number should be printed/displayed on each site
- 2.4 The method of fish identification employed must cause minimal suffering on fish both during the marking process and subsequently
- 2.5 Large sized fish can be tagged individually using PIT. However, for marking small sized fish external tagging would be safer
- 2.6 No matter, which type of tagging/ marking is opted, but it should be reasonably quick & simple to apply and easy to read/identify

Table 1. Log Book Format for Buildings ID Record

S.No	Building/Room /Hall/Net house	ID No	Area	Use	Year of Construction	Construction cost

Table 2. Log Book Format for Ponds/Tanks ID Record

S.No	Building/Room /Hall/Net house	ID No	Size	Year of Construction	Construction cost

Table 3. Log Book Format for Fish (brood stock) ID Record

S.No	Fish species	Source of procurement	Type of ID	ID No.	Special characteristics	Remarks

3. BMP-2: Keep breeding and seed rearing records for timely planning of marketing

The importance of breeding records is to measure the productive efficiency of the fish and to enable selection. For example, many farmers would like the fish which gives higher number of babies. Therefore, an accurate up-to-date breeding record of each individual is necessary (Table 4-5). The most important data in breeding records include: parentage (name or other identification of parents), fecundity, growth rate of offspring, etc.

Guiding factors for BMP- 2

- 3.1 Species wise, variety wise and sex wise number of brood stock used
- 3.2 Sex ratio of brood fish should also be recorded
- 3.3 Details of hormones used (type & dosage)
- 3.4 Breeding indices (species wise & variety wise fertilization, hatching and survival rates) be recorded carefully
- 3.5 Details of feeds used in nursery pond – type and quantity
- 3.6 Water quality in hatching and, nursery tanks/ponds.
- 3.7 Stocking density in the culture system- species & variety wise.
- 3.8 Duration rearing & harvesting details.
- 3.9 Survival percentage in the nursery system
- 3.10 Total value of harvested seed, etc. be recorded in log book for each and every cycle of breeding and nursery

Table-4 Log Book Format for Breeding Record

Date	Species (Fish)	Fish ID	Size of Fish	Sex ratio (F:M)	Hormone Used & Dosages	Fertilization (%)	Fecundity (%)	Hatching (%)	Total number of Larvae

Table-5 Log Book Format for Seed Rearing record

Date	Species (Fish)	Type of Rearing facility & ID No	Nos stocked per m ³	Type of Feed used	Feeding Rate	Rearing period	Size at harvest	Survival (%)	Total Fry (Nos)	Rate per Fry	Total Cost of harvested fry

4. BMP-3: Keep production records to monitor the production performance and profitability

These records are useful in measuring the performance of the fish. It contributes greatly to the economic appraisal of the enterprise. It can help farmers take decisions on investments, based on how many fish produce how much on the farm, so how much surplus can the family expects? The records can also be used by the whole sector to improve the genetics of the fish in the country, with specific focus on the production. Production records are kept of - fish products like eggs per fish per breeding cycle, phenotypic quality data. Production records are also necessary when farmers start selling the products and want to know of the balance available every day or every week or in a certain period. The intent of these records is to provide management with an accurate list of products sold (Table 6-8).

4. Guiding factors for BMP-3

- 4.1 Each expense or input item for pond/tank management and ornamental fish rearing should be described in as much detail as possible. e.g. pond management (fertilization, liming, etc.), stocked species and size.
- 4.2 Each item used, its date of application, its amount, its unit cost, and its total cost should be recorded.
- 4.3 As each pond is harvested, the following items should be recorded – date of harvest, species harvested, and amount harvested, and price received per unit.

Table-6 Log Book Format for Pond management Records

S.No	Pond/ Tank ID	Date	Type of Input	Quantity of input	Unit cost	Total cost (Rs)	Remarks

Table-7 Log Book Format for Stocking Records

S.No	Pond/ Tank ID	Date	Species	Source of Stocking	Stocking Rate	Average size	Total Nos	Unit cost	Total Cost

Table-8 Log Book Format for Harvesting Records

S. No	Pond/Tank ID	Date	Species harvested	Quantity Harvested (Nos)	Recovery/Survival (%)	Average size at harvest	Unit Price (Rs/fish)	Total Value (Rs)

5. BMP-4: Keep feeding records to save feed cost and availability of feed in store

Feeding records give information about the amount, type and quality of the feed. Feeding records can be used both for day to day management and adjustment of the feed ration (Table-9). It can be used for planning of activities related to feed conservation, etc.

Guiding factors for BMP-4

- 5.1 Feed costs are the single greatest expense in raising fish
- 5.2 A careful feeding schedule results in rapid and efficient fish growth
- 5.3 Carefully observe if there is any amount of feed leftover at the end of day
- 5.4 In case the fish is not consuming all the feed in reasonable time, the amount of feed should be reduced. It will avoid the wastage of feed and also help in maintaining water quality

Table-9 Log Book Format for Feeding Records

S.No	Pond/Tank ID	Date	Species (Fish)	Number of Fishes	Type of Feed	Feed Quantity	Balance Feed in Store	Feeding schedule	Unit Cost	Total cost

6. BMP-5: Keep disease and treatment records to prevent diseases outbreak

Disease and treatment records are necessary to keep track of the disease. It can guide to better management practices by leading the attention to repeated events or certain vulnerable groups of fish over time. It provides information about the health status of fish and it can help ensuring important treatments given at the right time. On basis of the disease and treatment records, success of interventions both for prevention and treatment can also be evaluated (Table-10). Disease and treatment records can for example involve: disease occurrence and date, treatments, etc.

Guiding factors for BMP-5

- 6.1 While precautions should be taken to prevent diseases, close monitoring and data recording are equally important.
- 6.2 Carry out a simple health inspection routine every day or weekly. Observe fish behavior, see if the fish is showing abnormal feeding or swimming pattern
- 6.3 If diseases symptoms are detected, seek assistance from the expert and record all the information in record log book.

Table-10 Log Book Format for Disease and Treatment Records

S.No	Pond/ Tank ID	Date	Species (Fish)	Diseases	Treatment	Cost of Treatment	Mortality (%)	Method of dead fish disposal

7. BMP-6: Keep weather condition and water quality records for improving production performance

It is suggested that a proper record of water quality (temperature, oxygen, ammonia, pH and nitrite levels) be kept (Table-11). It will facilitate to notice the problems well before they become threatening to the fish.

Guiding factors for BMP 6

- 7.1 Each ornamental fish production unit should have some means of evaluating water quality in order to be aware of the health of the fish culture system
- 7.2 Water quality analysis kits are popular for this purpose because they are relatively inexpensive
- 7.3 Monitor and record water quality daily that will help you to foresee the problems in advance and thereby avoid problems.

Table-11 Log Book Format for Water Quality monitoring Records

S.No	Pond/ Tank ID	Date	Weather Parameters			Water Quality Parameters						
			AirTemp (°C)	Rainfall (mm)	Humidity (%)	Temp (°C)	Colour	pH	DO (mg/l)	Hardness (mg/l)	Nitrate (mg/l)	Ammonia (mg/l)

8. BMP-7: Keep financial records to track the financial flow and economics of operations

The records of the costs and earnings related to the ornamental fish farming be kept for cash analysis and enterprise appraisal. In aquaculture, the most necessary records are simple overview over the cash flow that is the total economy of the farm: what comes in? and what do we buy? In addition to this, keeping records of the ornamental fish enterprises is an important part, because it can show whether it gives an income to the family or not. The financial records will also help farmer to see what they invest in it, and what it costs to produce it. Also in relation to the ornamental fish farm, an investment is more than expenditure? an investment hopefully enables and improves the production in the future. Economic records are of paramount interest in providing the farmer with information concerning the profitability. Moreover, economic records are required for tax purposes and for the purpose of getting loans & credits (Table 12-15).

Guiding factors for BMP- 7

- 8.1 Progress in ornamental fish business operation cannot be determined from year to year without an annual inventory.
- 8.2 Using the cash method for record keeping (i.e. keeping a record of receipts, expenses, and purchase cost of feed, fish sold during the year, plus depreciation schedules) will suffice to arrive at net income
- 8.3 Producers should maintain a balance sheet, a statement of owner equity, an income statement and cash flow statement.

Table-12 Log Book Format for Investment and Financing Record

S. No	Particulars	Year(s)										Total
		I	II	III	IV	V	VI	VII	VIII	IX	X	
1	Investment											
a	Land											
b	Buildings											
c	Equipment and Material											
d	Ponds/tanks											
e	Others											
	Booked Value											
2	Depreciation											
a	Buildings (5%)											
b	Equipment & Material (10%)											
c	Ponds/tanks (5%)											
d	Others (10%)											
	Total Depreciation											
3	Financing											
a	Equity											
b	Loans											
c	Repayment											
i	Principal											
ii	Interest											

Table-12 Log Book Format for Revenue (Per Month per Species) Record

Species	Sales price	Months												Total
		April	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	March	

Table-13 Log Book Format for Operational Cost (Per months) Record

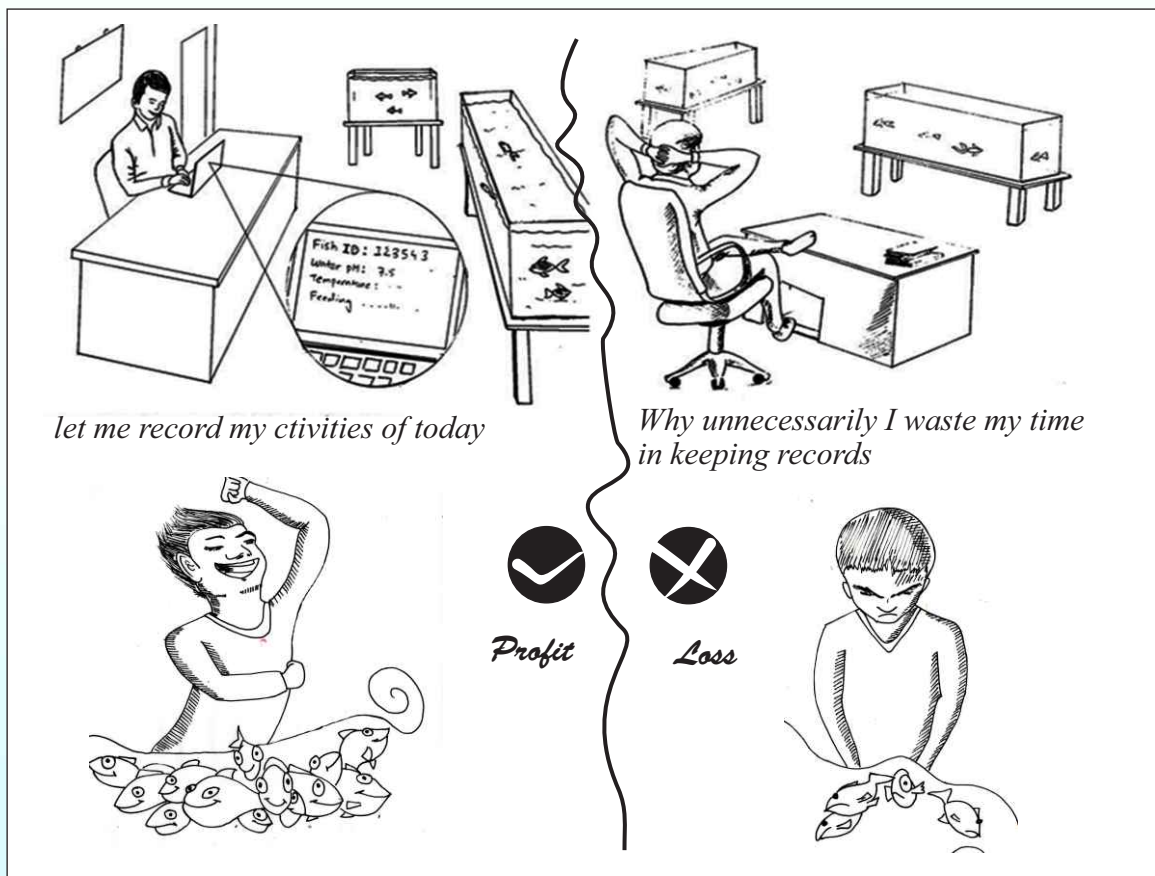
Particulars	Months												Total
	April	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	March	
Feed													
Medication													
Brood Stocks													
Electricity													
Water													
Labour / staff													
Others													
Contingency													
Total													

Table-14 Log Book Format for Cash flow Records

Cash Inflow				Cash Outflow				Profit (Cash Inflow- Cash Outflow)
Date	Source	Amount	Comments	Date	Source	Amount	Comments	
Total								

Table-15 Log Book Format for Financial Balance

Income and expenditure			
Expenditure (a)	Amount	Income (b)	Amount
To expenditure on labour, brooder, electricity, water, feed medicine, loan repayment, etc.		By fish sale	
To excess of income over expenditure (c=b-a)		Bank interest	
		Others	
Total			
Balance			
Liabilities	Amount	Assets	Amount
Opening balance		Cash in bank	
Add this year income(c)		Cash in hand	
		FDR	
Total			



9. Important Suggestions

The record keeping is important for efficient farm management. Using financial records and methodology will help you understand how and where your business is going. Record keeping and sound production and financial data interpretation will help you define the weakest links of your farm business operation and enable you to start corrective action plans. Keep the following guidelines in mind when implementing or reviewing your record keeping system.

- Keep it simple! If the record keeping system is unnecessarily complicated, you are more likely to make mistakes
- To identify the facilities and fish stock (brood stock) it is essential to keep the identification records
- Detailed of all inputs and harvest from each unit should be recorded to estimate the profitability
- To evaluate the breeding efficiency of farm brood stock detailed record of breeding indices (species wise & variety wise fertilization, hatching and survival rates) be recorded carefully
- Record the details of feeds used in nursery pond – type and quantity, water quality in hatching and, nursery tanks/ponds.
- Stocking density in the culture system- species & variety wise, rearing duration & harvesting details with value of harvested seed, etc. be recorded in log book for each and every cycle
- Carry out a simple health inspection routine every day or weekly, If diseases symptoms are detected, seek assistance from the expert and record all the information in record log book.
- The records of the costs and earnings related to the ornamental fish farming be kept for cash analysis and enterprise appraisal. The most necessary records are simple overview over the cash flow, that is, the total economy in the farming